

Modulated wetting

In both paper-based and thread-based microfluidics (Ballerini et al., 2011; Derda et al., 2009; Li et al., 2010; Martinez et al., 2008; Reches et al., 2010), liquid samples are transported using capillarity-driven wetting, hence no active mechanical pumping is required. Capillarity-driven flows have been applied previously in conventional microchannels (Parikesit et al., 2010a), where a constant pressure drop (i.e. the Young–Laplace pressure) between the channel inlet and the advancing liquid meniscus continuously drives

the liquid flow (Bruus, 2008). The capillarity-driven rise of liquids in narrow, hydrophilic, vertical tubes can be quantitatively predicted by the Washburn’s law (de Gennes et al., 2004), $z(t) = (R\gamma\cos\theta/2\eta)^{1/2} \times t^{1/2}$, where z is the liquid height in the tube, t is the liquid propagation time, γ is the surface tension, R is the tube radius, θ is the contact angle of the liquid at the tube wall, and η is the liquid viscosity. The Washburn’s law can also be applied to predict the capillarity-driven rise of liquids in porous media such as textiles (de Gennes et al., 2004). Here we hypothesize that the capillarity-driven wetting dynamics of textiles will be influenced by the textiles property, which includes the geometrical property (i.e. smooth vs. textured) and the material property (i.e. hydrophobic vs. hydrophilic).

To test this hypothesis, we set our samples in a vertical position and dipped them into a reservoir containing a dark-green food dye (Koepoe-Koepoe, Jakarta, Indonesia, composition: Tartrazine CI 19140, Brilliant Blue CI 42090), such that capillarity-driven wetting occurs on the samples. The wetting process on the samples is monitored for three different wetting directions (see Figure 1) using a digital webcam (PCC8020A IzyCam, ProLink, Jakarta, Indonesia), after which all sets of the digital images are processed and analyzed using the software ImageJ (<http://www.rsweb.nih.gov/ij/>) to determine the capillarity-driven flow dynamics on the samples.

Figure 1 shows snapshot images taken during the wetting of three different types of samples: (a) paper sheets (HVS, 80gr, PaperOne, Jakarta, Indonesia), (b) grid clothes, and (c) seat cover clothes. The red, green, and blue arrows indicate the directions along which we monitor the wetting process. The capillarity dynamics along the red, green, and blue arrows are qualitatively similar for the paper sheets (Figure 1(a)) and the grid clothes (Figure 1(b)) because both of these samples have spatial distribution of texture that is homogeneous. Meanwhile, for the seat cover clothes (Figure 1(c)), the capillarity flow along the blue arrow is significantly faster than the flow along the red and green arrows, due to the heterogeneous spatial distribution of the texture.

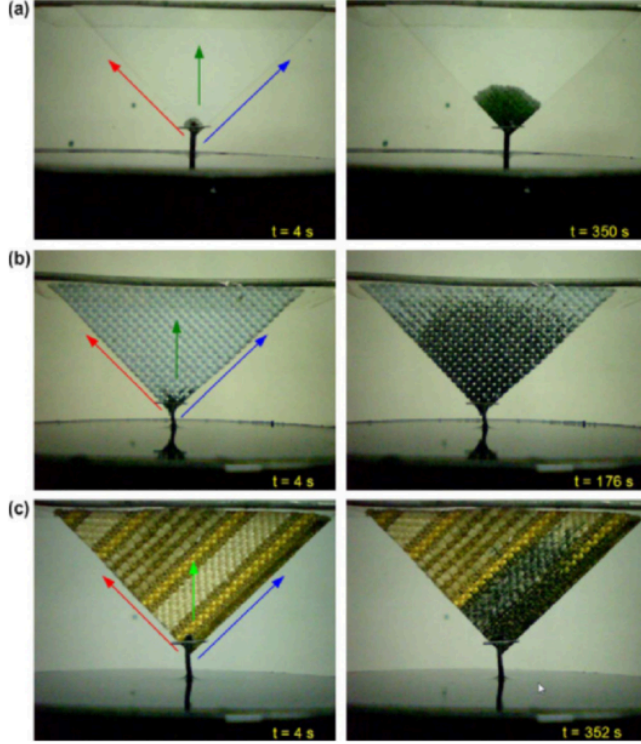
The measured capillarity-flow dynamics in all the samples are quantitatively shown in Figure 2, where we plot the values of $z(t)$, with the solid fitting lines represent the fitting curves. Here we see that, the flow dynamics in the paper sheets (Figures 1(a) and 2(a)), which serve as the control experiment in our study, are well predicted by the Washburn's law. As for the grid clothes (Figures 1(b) and 2(b)), even though the flow dynamics still qualitatively agree with the Washburn's law, the quantitative values differ significantly, particularly on the fitting powers of t (~ 0.8 as opposed to

~ 0.5). More interesting phenomena are observed with the seat cover clothes (Figures 1(c) and 2(c)), where quantitative variations of the fitting powers of t become more prominent for each different directions of flow.

In order to further analyze the phenomenon observed with the seat cover clothes, we examine the wettability variation among all types of threads in the clothes, with each measurement repeated for 12 different specimens. The data in Figure 3 show the wettability variation among the different types of threads, which modulates the capillarity-driven flow dynamics in the seat cover clothes (Figures 1(c) and 2(c)). All the data in Figures 1–3 confirm that the capillarity-driven wetting dynamics of textiles are significantly influenced by both the geometrical property (i.e. smooth vs. textured) and the material property (i.e. hydrophobic vs. hydrophilic) of the textiles.

Liquid mixing

Based on the results obtained in the previous section, in which the textile texture was shown to modulate the capillarity-driven wetting dynamics on textile samples, we then investigate how the textile texture influences the process of liquid mixing in textile-based microfluidic devices. In general, mixing in the microscopic scale cannot be performed using inertial forces, due to the small Reynolds numbers found in microfluidic systems (Bruus, 2008). In order to solve this, previous researchers have used T-mixer devices, where microfluidic mixing of two dyed liquids was performed based on inter-diffusion between the two types of dye molecules (Osborn et al., 2010). For our experiments, we used blue and yellow printer inks (Acacia, Bandung, Indonesia) and Gabardine cotton clothes (see the texture in Figure 4(a), taken with a webcam-based microscope; Parikesit et al., 2010b). The measured diffusivity constants of the blue and yellow inks on the Gabardine clothes are 1.49×10^{-5} and $5.24 \times 10^{-6} \text{ m}^2/\text{s}$, respectively. During the mixing experiments, the T-mixer device is connected to two different inlet reservoirs using hydrophilic elastic bands. The reservoirs' heights are varied during the experiment to investigate the influence of convection on the liquid mixing quality in the microfluidic devices. In order to study the effects of textile orientations, the aligning of the Gabardine texture with respect to the flow direction is also varied (i.e. 0° and 45°). The geometry of the device is illustrated in Figure 4(b). Figure 4(c) shows an example of data images captured by the webcam, showing diffusion-driven mixing between blue and yellow printer inks on the textile device; also shown next to the device is capillarity-driven wetting patterns of blue and yellow printer inks on the Gabardine clothes.



We analyze the mixing quality in three different locations along the microfluidic device, i.e. at upstream, at the middle, and at downstream. The mixing quality is quantitatively defined as follows:

$$Q_m = 1 - \frac{\sum_{j=1}^{20} \frac{|h_j - 0.5|}{0.5}}{\sum_{i=1}^{20} i}, \quad (1)$$

where Q_m is the mixing quality, j (from 1 to 20) is the measurement points in the analysis (taken either at upstream, at the middle, or at downstream), and h_j is the measured 'hue ratio' in each measurement point, analyzed using a customized colorimetry method developed in ImageJ. In this colorimetry method, the 'hue ratio' is the ratio between the hue value of the

sampled color (in each measurement point) and the hue values of unmixed blue and yellow inks, respectively. Hence unmixed blue inks would result in $h_{\text{blue}}=1$ and $h_{\text{yellow}}=0$, while unmixed yellow inks would result in $h_{\text{blue}}=0$ and $h_{\text{yellow}}=1$. The value of 0.5 used in Equation (1) corresponds to the condition of perfect mixing, where $h_{\text{blue}}=h_{\text{yellow}}=0.5$. Both unmixed blue inks and unmixed yellow inks would therefore result in $Q_m=0$, while perfectly mixed inks lead to $Q_m=1$.

Table 1 comprehensively shows the measured values of Q_m as functions of: the location of the analysis (i.e. at upstream, at the middle, and at downstream); the height of the blue ink reservoir ($H_{\text{blue}}=-3.5, 0$ and 7 cm); the height of the yellow ink reservoir ($H_{\text{yellow}}=-3.5$ and 0 cm); and the aligning of the Gabardine texture with respect to the flow direction (i.e. 0° and 45°).

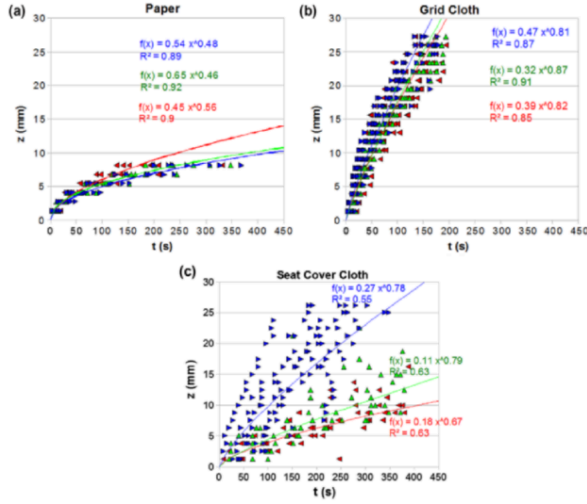


Figure 2. Capillarity-driven flow dynamics of the three samples shown in Figure 1.

Based on the data shown in Table 1, we can conclude several things. First, the mixing quality at the downstream is generally higher than at the middle and at the upstream locations, i.e. $Q_{m[\text{downstream}]} \gg Q_{m[\text{middle}]} \gg Q_{m[\text{upstream}]}$. Second, the textile texture does not significantly influence the mixing quality, except when $H_{\text{blue}} = -3.5$ cm and $H_{\text{yellow}} = -3.5$ cm, i.e. when the diffusion effects strongly dominate the convection effects (NB. negative height of the reservoirs means that the capillarity pressure that drives the wetting of the textile is retarded by the opposing hydrostatic pressure toward the inlet reservoirs.) Third, when the textile texture *does* influence the mixing quality, we have a general trend that $Q_{m[\text{alignment}=45^\circ]} \gg Q_{m[\text{alignment}=0^\circ]}$, i.e. the diagonal threads' orientation (45°) allows the blue and yellow dye molecules to experience a better inter-diffusion.

Liquid sorting

In this section, we investigate how the textile texture influences the process of liquid sorting in textile-based

microfluidic devices. Similar to liquid mixing, liquid sorting in microscale also depends mainly on diffusion (Osborn et al., 2010). In order to sort one type of dye molecules out of a mixture, the diffusivity of that particular type of molecules should be sufficiently higher than the other molecules comprising the mixture. For our experiments, we used blue printer ink (Acaciana, Bandung, Indonesia), yellow food dye (Koe-poe-Koe-poe, Jakarta, Indonesia, composition: Tartrazine CI 19140), and Oxford cotton clothes (see the texture in Figure 5(a), taken using a webcam-based microscope; Parikesit et al., 2010b). The measured diffusivity constants of the blue printer ink and the yellow food dye on the Oxford clothes are 2.48×10^{-6} and 1.81×10^{-6} m²/s, respectively. A green liquid mixture is made out of the blue ink and yellow food dye (with 1:1 volume ratio) for this experiment.

We design our liquid-sorting device, which has a 15×2 cm² rectangular shape, based on the sorting method of chromatography. A droplet of the green liquid mixture was added at a certain point on the

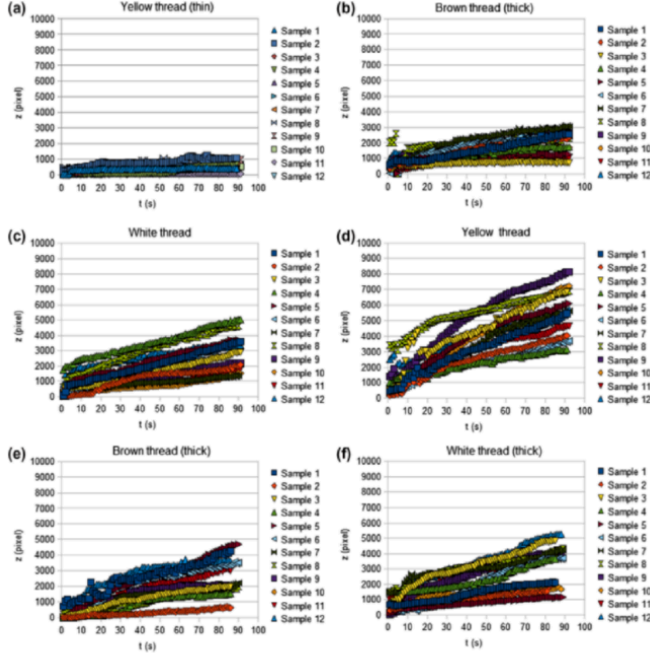


Figure 3. Capillarity-driven wetting dynamics of each type of threads from the seat cover clothes shown in Figure 1(c).

device, after which a droplet of water was added at another point on the device (see Figure 5(b)). We vary the volumes of both the green mixture ($V_{\text{blue}}=47.5, 95$ and $285\mu\text{L}$) and the added water ($V_{\text{water}}=190$ and $380\mu\text{L}$) that are dropped on the device. In order to investigate the texture effect on the sorting quality, we also vary the alignment of the two dropping points (i.e. 0° and 45°) with respect to the textile-based sorting device.

Figure 5(c) shows an example of data images captured by the webcam (note: the liquids flow from the right side towards the left side of the image), demonstrating the device's ability to sort blue printer ink out of the green mixture (also shown in the Figure 5(c)) are samples of unmixed blue printer ink, unmixed yellow

food dye, as well as the green mixture on the Oxford clothes). We analyze the quality of sorting at three different locations on the textile-based device, i.e. at the north-side, at the middle and at the south-side. The sorting quality, Q_s , is quantitatively defined as follows:

$$Q_s = \max(h_{\text{blue}}) - \min(h_{\text{yellow}}), \quad (2)$$

such that unmixed blue inks ($h_{\text{blue}}=1$ and $h_{\text{yellow}}=0$) would result in $Q_s=1$ (i.e. ideal sorting), while perfectly mixed inks ($h_{\text{blue}}=h_{\text{yellow}}=0.5$) lead to $Q_s=0$ (i.e. no sorting has occurred).

The data shown in Table 2 allow us to draw several conclusions. First, the sorting quality variation

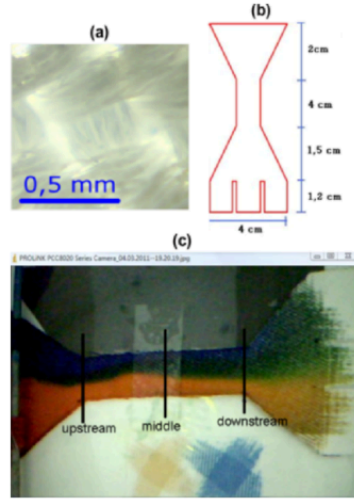


Figure 4. (a) Texture of the Gabardine clothes used for the liquid mixing experiments. (b) The geometry of the mixing device. (c) An example of data images captured by the webcam, showing diffusion-driven mixing between blue and yellow printer inks on the textile device; also shown next to the device is capillarity-driven wetting patterns of blue and yellow printer inks on the Gabardine clothes.

between the various analysis locations depends on the texture alignment. For a texture alignment of 45° , the sorting quality at the north-side is generally higher than at the middle and at the south-side location, i.e. $Q_{s[\text{north-side}]} \gg Q_{s[\text{middle}]} \gg Q_{s[\text{south-side}]}$. For a texture alignment of 0° , the sorting quality at the north-side is generally similar to the sorting quality at the middle and at the south-side location, i.e. $Q_{s[\text{north-side}]} = Q_{s[\text{middle}]} = Q_{s[\text{south-side}]}$. This phenomenon can be explained simply by the direction of the flow, as determined by the texture alignment. A texture alignment of 45° leads to flows that are directed specifically towards the north-side of the device, while a texture alignment value of 05° leads to flows that are directed generally towards all parts at the end of the device. Second, the ratio between V_{green} and V_{water} determines the values of Q_s ; in general, $Q_s[V_{\text{green}} \ll V_{\text{water}}] \gg Q_s[V_{\text{green}} \approx V_{\text{water}}]$. This indicates that the dye molecules experience higher diffusivity in a wetted cloth than in a dry cloth. Third, the effect of textile align-

ment depends on the ratio between V_{green} and V_{water} . In general, for $V_{\text{green}} \ll V_{\text{water}}$ we have $Q_{s[\text{alignment} = 45^\circ]} \gg Q_{s[\text{alignment} = 0^\circ]}$, while for $V_{\text{green}} \approx V_{\text{water}}$, $Q_{s[\text{alignment} = 45^\circ]} = Q_{s[\text{alignment} = 0^\circ]}$.

Biopis 20-04-2026

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Lab on a chip → Lab yg besar direduksi mjkd miniatur sebesar uang koin (?) Intinya Lab besar direduksi mjkd kecil

Microfluidics → ilmu mekflu versi mikro

Mekflu = mikro

- Laminar

⊙ Reynolds kecil

⊙ gaya viskos yg dominan

vs makro

- turbulent

⊙ Reynolds besar

⊙ gaya inersia nya yg dominan

kecepatan terkait ukuran → viskositas

Reynolds → Reynolds = gaya inersia / gaya viskos

- capillary-driven filling vs

⊙ Fenomena air yg masuk ke tgl kosong sprt tisu

⊙ Hidrofilik vs Hidrofobik

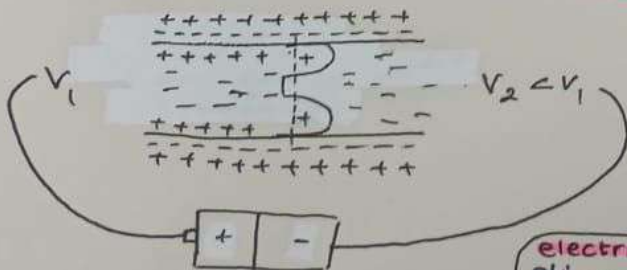
menyerap air ←

menolak air

pump-driven filling

- electrically-driven pumping vs

pressure-driven pumping



$$\Delta P \propto \frac{1}{\text{ukuran}}$$

electrically-driven pumping: aliran di dinding lbh cepet dari yg ditengah beda dr skala makro

⊙ Pipa tergantung ukuran: pressure-driven pumping jika ukurnya gede lbh condong ke turbulent sbktk jika ukuran kecil akan lbh condong ke laminar begitupun y/ electrically-driven pumping

⊙ Pipa disebut electrically / pressure trngntng ujung-2 dikasih P atau V.

⊙ Reynolds = ukur kecil → gaya viskos dominan → Reynolds kecil
= ukur besar → gaya inersia dominan → Reynolds besar

Catatan Biofisika Senin 4/05/20.

* Review 2 Minggu Lalu

→ Konsep Dasar : Lab on a chip (Mikrofluida)

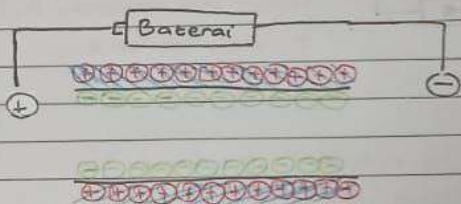
- ① Dalam sistem ini tak menggunakan pompa mekanik. Sbg gantinya dpt menggunakan gaya listrik $\forall$ menggerakkan fluida. Karena pompa sulit dikalakan ke ukn mikrometer.

distribusi kece-
pakan

- ① Jika pompa biasa alrn berbentuk parabolik (rec. max dite ngah.

- ① Pd penggerak listrik (alrn cenderung rata / plug flow)

* Fenomena penggerak berbasis listrik



3 Fenomena Utama :

Elektro Osmosis

- ① B'tkaitan dg p'grtn slrh vol. cairan di dlm slrn mikro g memiliki muatan p'mukaan
- ① Prinsip : Dinding Saluran (biasax silika / kaca) cenderung b'mu atan (-).
- ① Mekanisme : Saat medan listrik dibe kan, ion (+) di lapisan EDL bgkn menuju ke arah (-) sambil menyeret molekul air disekitarnya
- ① Hasil : Cairan mengalir scr kslrh

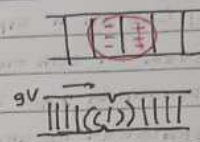
Electrophoresis

- ① Fokus pd pergerakan partikel / benda di dlm cairan
- ① Prinsip : Partikel g punya muatan atm b'grt jk diletakkan dlm medan listrik apapun muatannya.
- ① Mekanisme : Partikel muatan (+) ditarik ke katoda (-) dan sebaliknya
- ① Kegunaan : Memisahkan molekul b'dsrtn ukuran t muatannya.

Dielectrophoresis

- ① Fenomena g lbh trom Pleks (bs menggerak kan benda g netral)
- ① Prinsip : Menggunakan medan listrik g tak seragam.
- ① Mekanisme : Benda netral mgkn pola risasi di dlm me dan listrik g jk tak merata partikel akan terdorong ke arah medan g lbh kuat / lemah t'ntang sifatnya.
- ① Kondisi :
 1. Medan seragam, par tikel hax diam / b'ge tar b'gaya total = 0
 2. Medan tak seragam, Trjd p'grtn t'm p'bedaan kekuatan tarikan

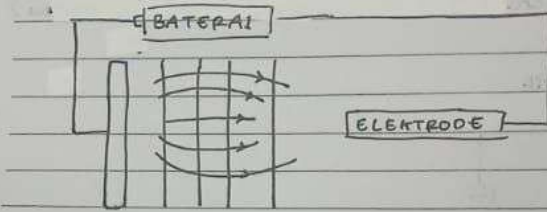
Electrophoresis dan Dielectrophoresis



medan listrik kuat

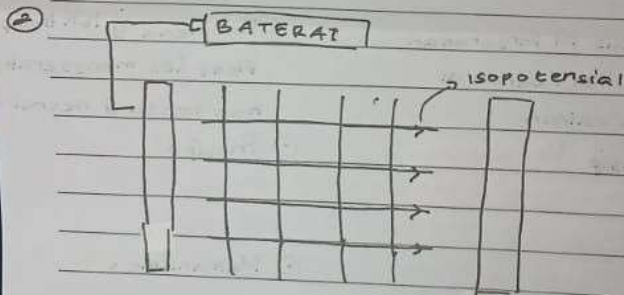
= kalo ada medan listrik yg mengecil

① Medan Listrik Tak seragam



Kondisi yg mrptn syarat mutlak t'jadi Dielectrophoresis yg t'pola r'isasi krn ketuatan medan di 1 sisi partikel yg lbh kuat.

Digunakan " untuk memisahkan sel mati dari sel hidup krn sel mati memiliki sifat listrik (dielektrik) yg berbeda.



Partikel b' muatan akan b'grk scr lurus, namun " partikel dg muatan netral tdk atrn b'ndh t'q krn gaya pada ke-2 sisi y t'polarisasi akan saling meniadakan

No.

Date

Biologi

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Elektroosmosis

↳ adalah pergerakan fluida melalui medium berpori atau kanal kecil akibat adanya medan listrik eksternal.



Prinsip kerja

- Penerapan medan listrik : 2 elektroda, anoda (positif) + katoda (negatif) diterapkan ke dlm tanah atau pd ujung medium
- Muatan permukaan tanah : Partikel tanah lempung secara alami bermuatan negatif ketika basah
- Pembentukan Lapisan Ganda : menetralkan muatan negatif tanah, kation (ion positif) dlm air pori akan berkumpul di dekat permukaan partikel tanah membentuk "lapisan ganda"
- Migrasi Air : ketika arus DC dialirkan, kation yang bermuatan positif di lapisan ganda akan bergerak menuju katoda.
- Aliran massa air : Kation⁺ yg bergerak membawa molekul air yang bersifat polar bersamanya. Proses ini membuat air mengalir dari anoda menuju katoda.
- Deswatering : Air yang terakumulasi di katoda dapat dipompa keluar, mengurangi kadar air dalam tanah.

Persamaan elektroosmosis : $u_{eo} = \frac{e\zeta}{4\pi\eta} E$

Aplikasi : • microfluidics / lab-on-chip

• pemisahan sel

• soil remediation

• transport sensor tanpa pompa mekanik

Elektroforesis

↳ adalah pergerakan partikel bermuatan dalam suatu medium akibat medan listrik

Prinsip elektroforesis

- pemisahan komponen molekul bermuatan (sperti DNA, RNA atau protein) dalam medium penyangga berdasarkan perbedaan laju migrasi saat dialiri listrik searah (DC)

Komponen Utama

- Pergerakan dalam medan listrik : Molekul bermuatan negatif (seperti DNA) akan bermigrasi dari kutub negatif (katoda) ke kutub positif (anoda).
- Ukuran dan bentuk molekul : Partikel yang lebih kecil & padat akan bergerak lebih cepat & lebih jauh melalui pori-pori gel dibandingkan partikel lebih besar.
- Matriks Gel : Gel bertindak sebagai saringan yg memisahkan molekul berdasarkan ukurannya.
- Lantian buffer : Lantian elektrolit yang berfungsi menghantarkan arus listrik dan menjaga pH tetap stabil.

Aplikasi : • Analisis DNA / RNA • Identifikasi protein
• Diagnostik medis • Forensik

Dielektroforesis

LS adalah pergerakan netral (tidak bermuatan) dalam medan listrik tak seragam akibat polarisasi listrik.

Prinsip

- Medan Listrik Non-Uniform (Inhomogen) : memerlukan medan listrik yg tidak seragam biasanya dihasilkan oleh mikroelektroda dengan geometri tertentu.
- Polarisasi Partikel :
 - Interfacial Polarization : gaya pd kedua ujung dipol tidak sama besar
 - Net force : menghasilkan gaya neto km perbedaan gaya.
- Arah Gaya DEP :
 - Positive : jika partikel lebih polarisabel drp mediumnya, partikel ditarik menuju area dgn intensitas medan listrik tertinggi
 - Negative : jika partikel kurang polarisabel drp mediumnya, partikel didorong menjauhi area intensitas medan tertinggi menuju area rendah

Faktor yang mempengaruhi :

- Polarizabilitas partikel
- konduktivitas medium
- Frekuensi medan listrik (AC)
- ukuran partikel

Mekanisme Polarisasi :

- Maxwell-Wagner Polarization
- Orientation polarization

Aplikasi :

- Manipulasi sel dan DNA
- Microfluidic particle sorting
- Lab-on-chip
- biosensor